ICT171 Reflective Journal

MU BRG-ISEA

Introduction to Server Environments and Architectures

Item	Details
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Additional Server Service	Docker Containerisation
GitHub Repository Link	https://github.com/Selkhor/ICT171-Lab-Journal
Video URL Link / uploaded filename	https://m365kloud.sharepoint.com/sites/LEARN/_layouts/15/stream.aspx?id=%2Fsites%2FLEARN%2FShared%20Documents%2Fe%2DKAPLAN%2FMU%2DISEA%2Fstudent%2FVideoUpload%2FCT0371957%2DKhorWounHang%2DAssignmentISEA%2Emp4&referrer=StreamWebApp%2EWeb&referrerScenario=AddressBarCopied%2Eview%2E883f9971%2D9d73%2D4930%2Daa2c%2D77d36715fdd8 CT0371957-KhorWounHang-AssignmentISEA.MP4
GitHub pages saved as PDF	CT0371957-KhorWounHang-AssignmentISEA-GIT01.pdf

Word DOCX file uploaded on LMS	CT0371957-KhorWounHang-AssignmentISEA.docx

Submission Date:17 June 2026

1. Introduction

This reflective journal documents my learning journey throughout ICT171. The unit introduced me to Linux system administration, cloud computing, scripting, automation, DNS management, web security, and containerisation technologies. All practical activities were completed using Ubuntu 22.04 ARM64 running on UTM virtualisation software on a MacBook M1. Cloud-based exercises were completed using Amazon Web Services (AWS) EC2.

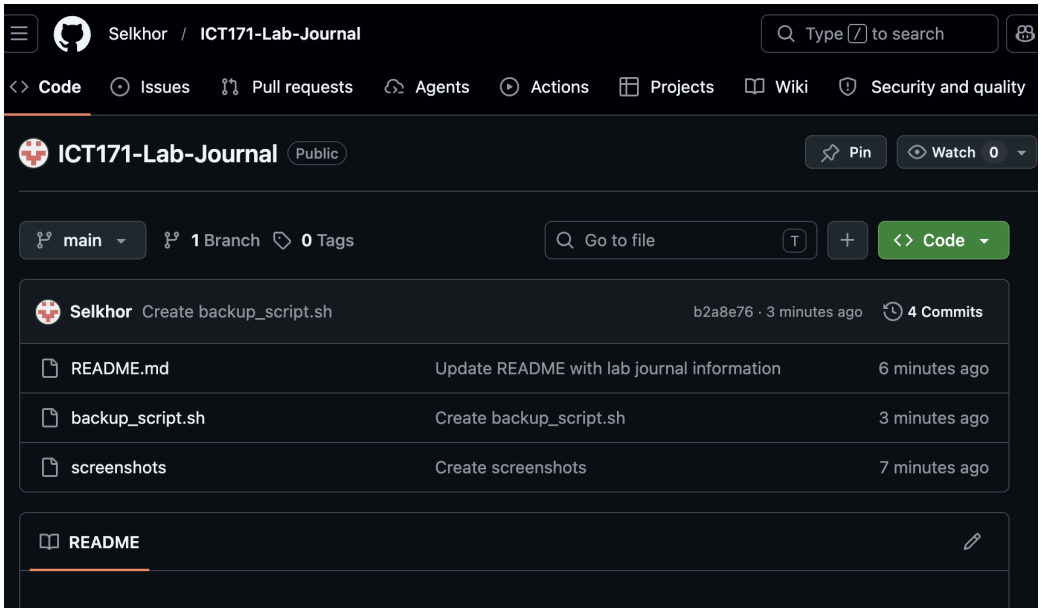
Before starting ICT171, I knew very little about Linux and had no real experience with cloud computing. At the beginning, many of the commands and concepts seemed unfamiliar. As the semester progressed, I gradually became more comfortable working with Linux, AWS, Bash scripting, DNS, HTTPS, and Docker. This journal describes the activities I completed, the challenges I faced, the solutions I found, and how these skills relate to real-world IT environments.

2. Linux Environment Setup

2.1 GitHub Repository

At the beginning of the semester, I created a GitHub repository called ICT171-Lab-Journal to store screenshots, scripts, notes, and laboratory evidence.
Using GitHub allowed me to organise my work systematically and maintain version control throughout the semester. Every major task and script was documented and uploaded to the repository.

Screenshot 1

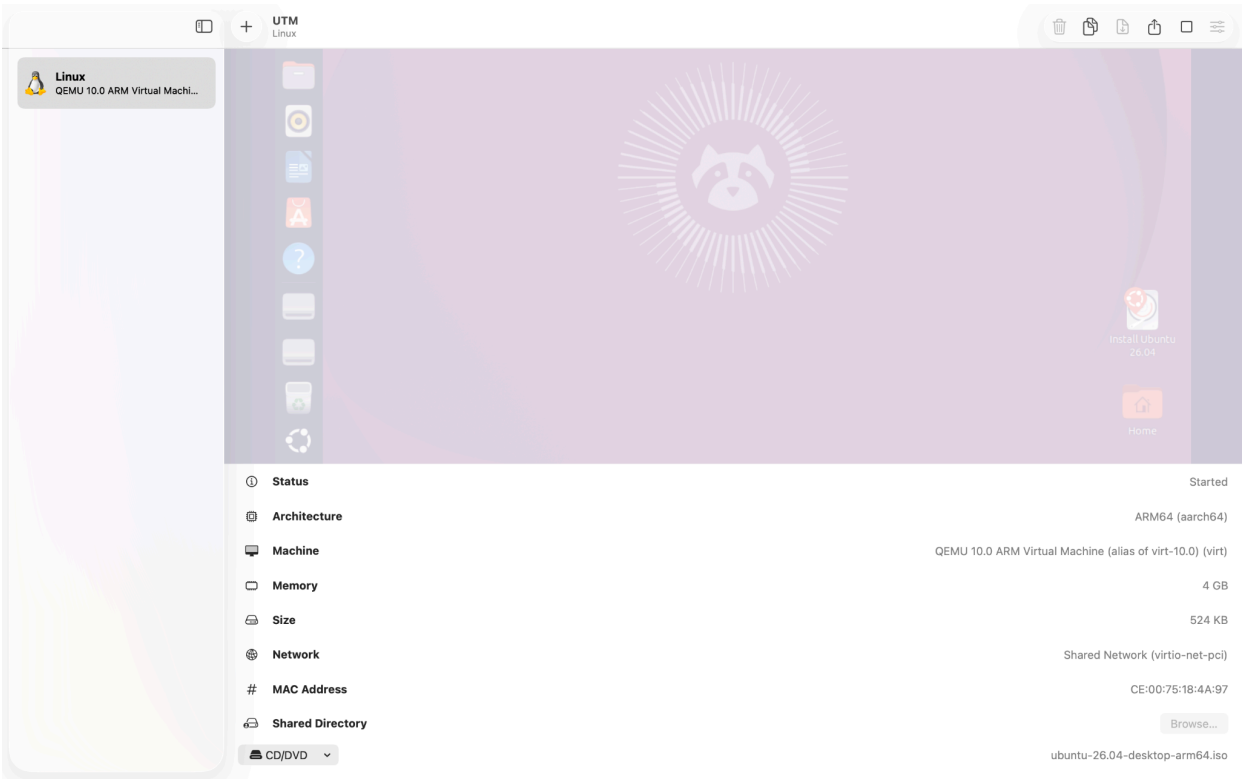


Reflection:I thought GitHub was just for code. Then I made my own repo for this class. Called it ICT171-Lab-Journal. Threw all my screenshots, scripts and notes in there. After each lab I just uploaded everything. One time I messed up a script. Went back to an old version and fixed it. Saved me. At work I guess teams would do the same with config files.

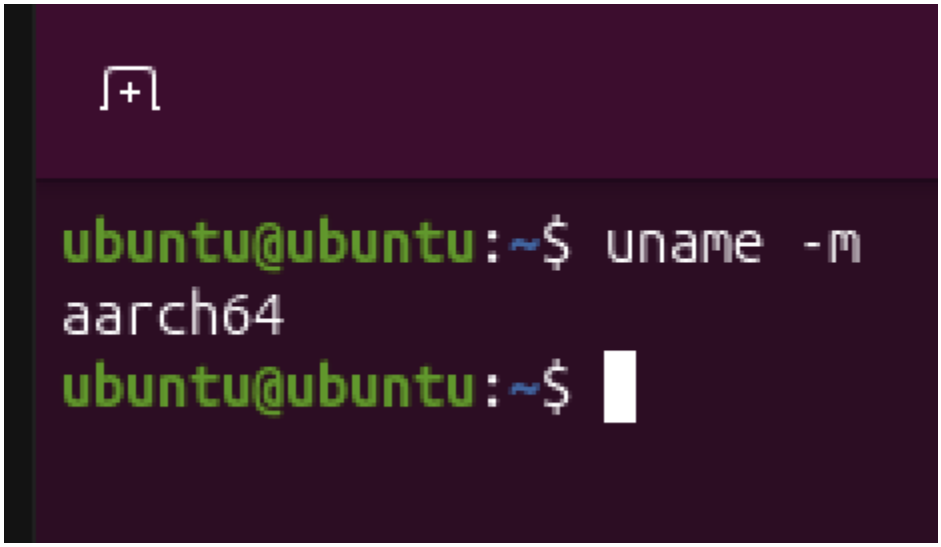
2.2 Installing Ubuntu on Apple Silicon

Because I use a MacBook M1, traditional virtualisation platforms such as VirtualBox were unsuitable. Instead, I used UTM to create an Ubuntu 22.04 ARM64 virtual machine.
Initially, I downloaded the x86_64 version of Ubuntu, which caused compatibility problems. After identifying the issue, I downloaded the ARM64 version and successfully installed Ubuntu.
Verification was performed using:
bash
uname -m
which returned:
bash
aarch64
confirming the operating system was running on ARM64 architecture.

Screenshot 2



Screenshot 3



Reflection: I made a mistake at the start: I downloaded the x86_64 version of Ubuntu for my M1 Mac. When I tried to install it, nothing worked. After some searching online, I realised the problem — my Mac uses ARM64 architecture, not x86. That's when I found UTM and downloaded the correct ARM64 ISO. Later in the semester, this experience paid off when I launched AWS Graviton instances. They use the same ARM architecture, so I already understood why they exist and how they're different from regular x86 servers.

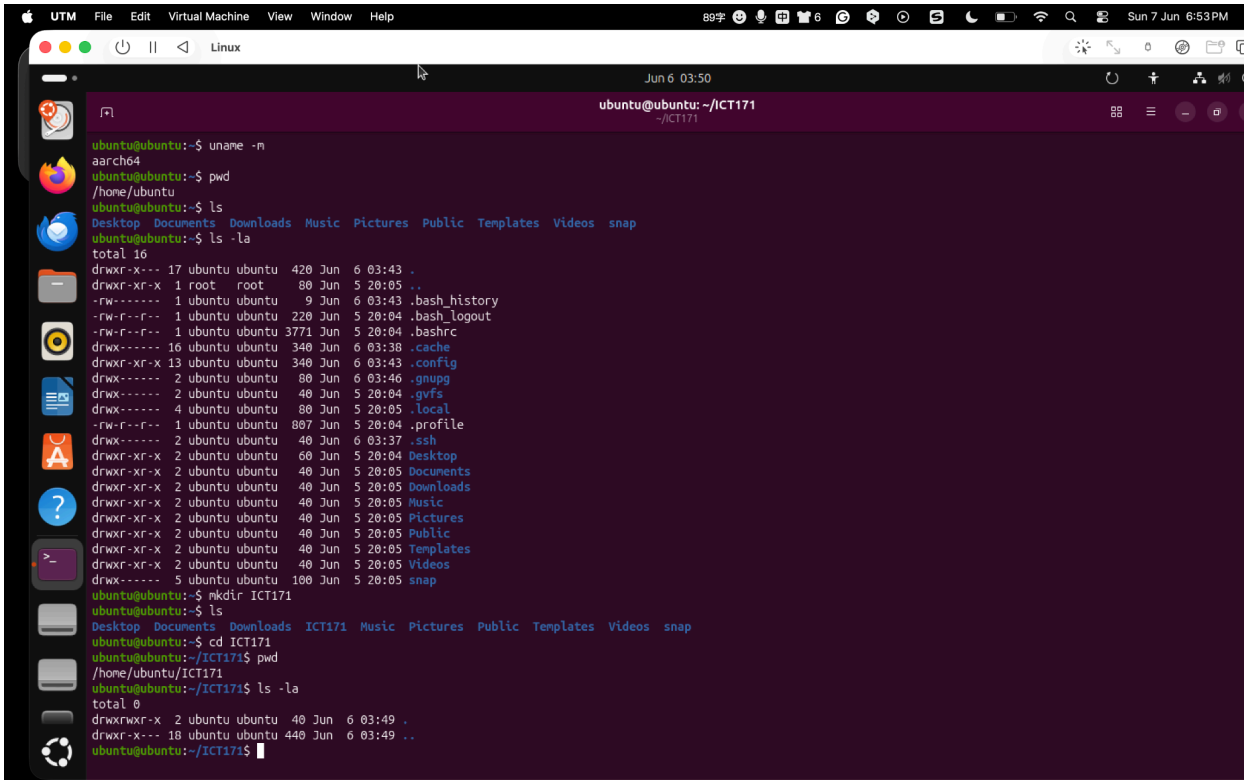
2.3 Ubuntu Command Line Familiarisation

I learned a variety of Linux commands including:

Command	Purpose
ls -la	List all files with permissions
cd, pwd	Navigate directories
mkdir, touch	Create directories and files
cp, mv, rm	Copy, move, delete files
grep, find	Search within files and directories
sudo systemctl	Manage services
ip a, ping	Network configuration and testing

I also installed software using the APT package manager and compiled simple programs using GCC.

Screenshot 4



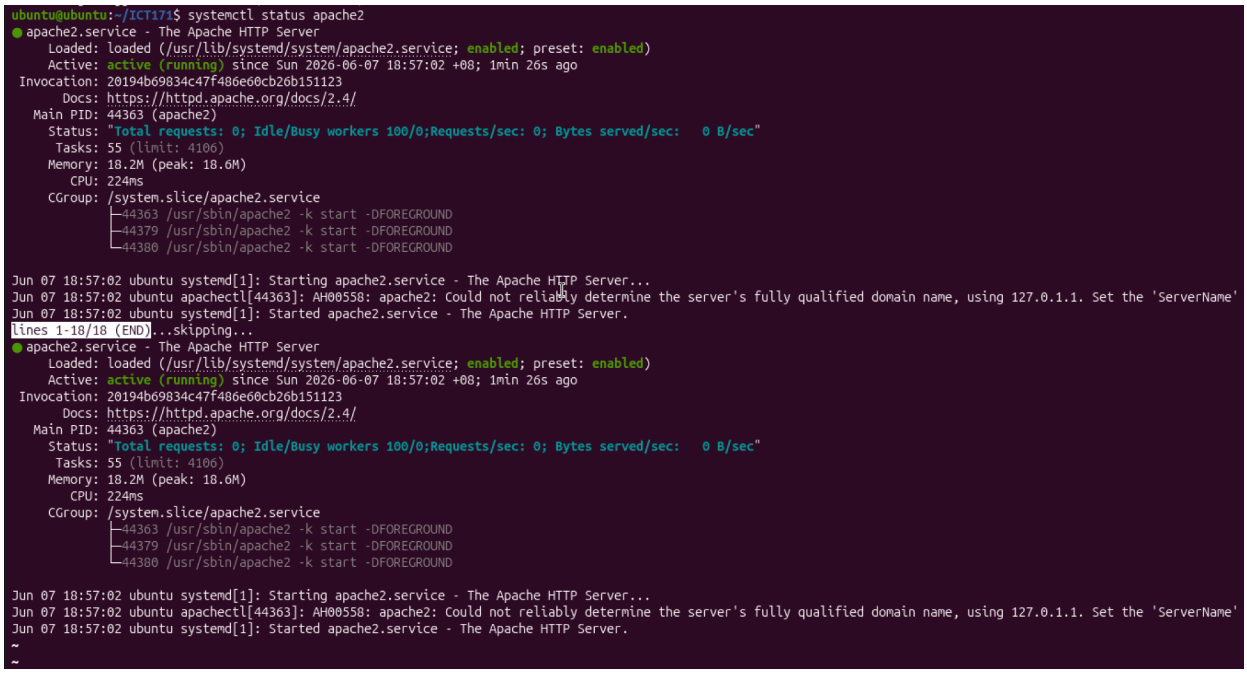
Reflection: At first, using the command line felt uncomfortable because I was more familiar with graphical user interfaces. However, after repeated practice I realised that command-line tools offer greater efficiency, flexibility, and automation potential. These commands became the foundation for every subsequent activity in the unit.

3. Linux Services, Permissions and File Searching

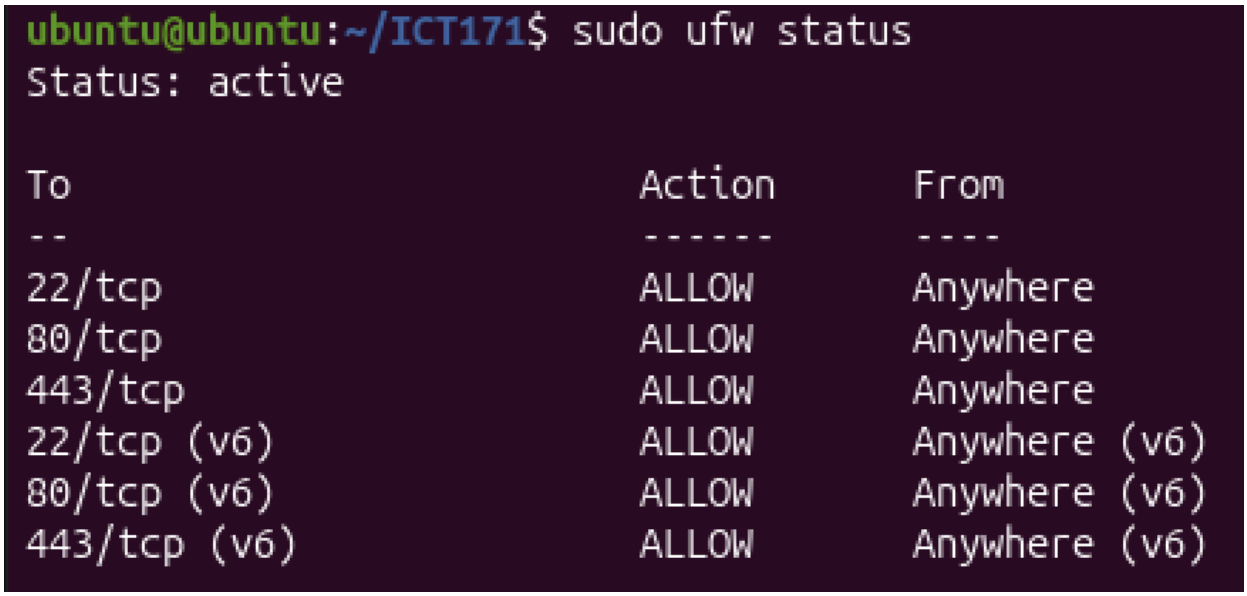
3.1 Linux Services

Apache was installed and managed using:
bash
sudo apt install apache2 -y
sudo systemctl start apache2
sudo systemctl enable apache2
Firewall rules were configured using UFW to allow SSH, HTTP, and HTTPS traffic.
bash
sudo ufw allow 22,80,443/tcp
sudo ufw enable
SSH key-based authentication was also tested.

Screenshot 5



Screenshot 6



Reflection: I didn't really think about how websites stay online before this lab. After installing Apache and managing it with systemctl, I started to understand how Linux services run in the background. I also ran into situations where Apache was running but still couldn't be accessed because of firewall settings. That helped me realise that server administration is more than just installing software — networking and security matter too.

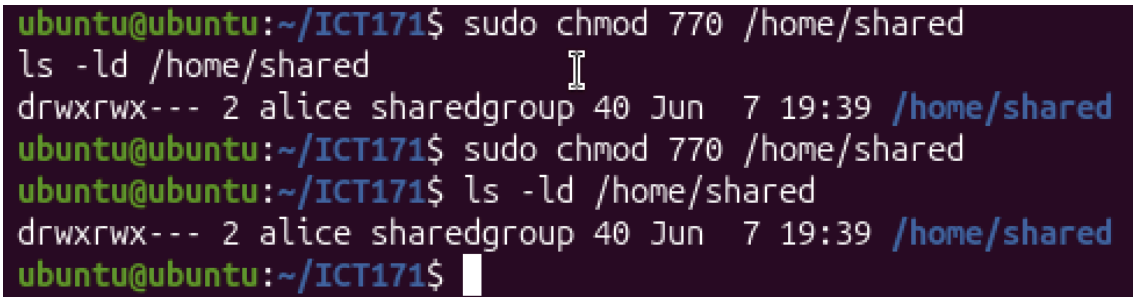
3.2 Linux Permissions

User accounts and groups were created to explore Linux access control mechanisms.

User	Access Required	Command
alice	Read, Write, Execute (rwx)	chmod 770
bob	Read and Execute only (r-x)	chmod 750
mallory	No access (---)	chmod 000

Permissions were modified using chmod, chown, and chgrp.

Screenshot 7

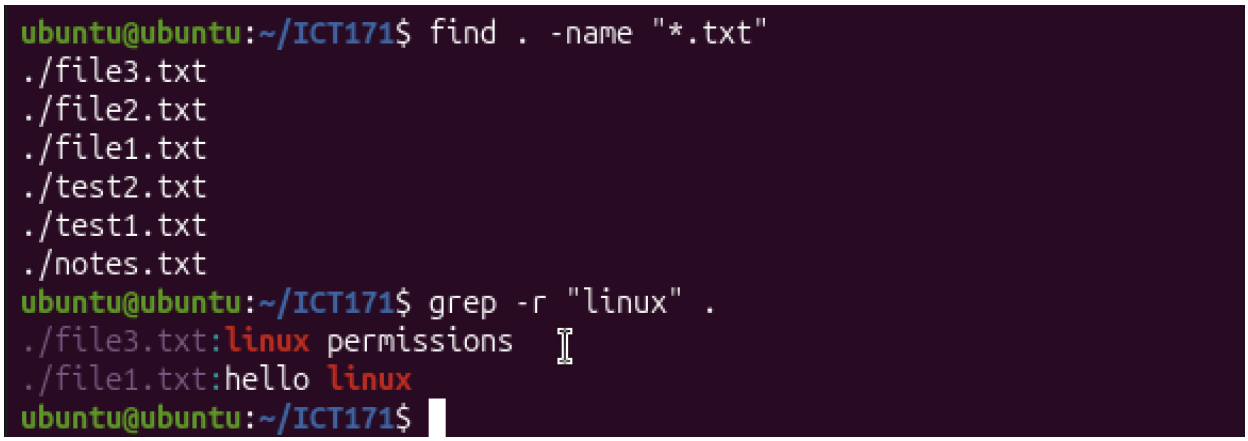


Reflection: I'll admit — at first I just used chmod 777 whenever something said 'permission denied'. It worked, so I didn't think about it. But then I created users alice, bob, and mallory and tried to give them different access levels. I saw that 777 gives everyone full access to everything. That's dangerous. Now I use 750 or 770 and only give exactly what each user needs. This 'least privilege' idea makes a lot of sense for keeping systems secure.

3.3 File Searching and Analysis

The find and grep commands were used to search large datasets and locate specific files.
Examples included:
bash
find ./Gutenberg -name "*.txt"
grep -r "verdigris" ./Gutenberg
These tools enabled efficient searching of thousands of files. I successfully located answers to lab questions including counting keyword occurrences and identifying authors of specific files.

Screenshot 8



Reflection: Before using find and grep, I would normally search for files manually. After working with larger datasets, I realised how much faster command-line tools are. Being able to locate files and specific keywords within seconds showed me why these tools are so useful for system administration and troubleshooting.

4. Cloud Infrastructure and TCO Analysis

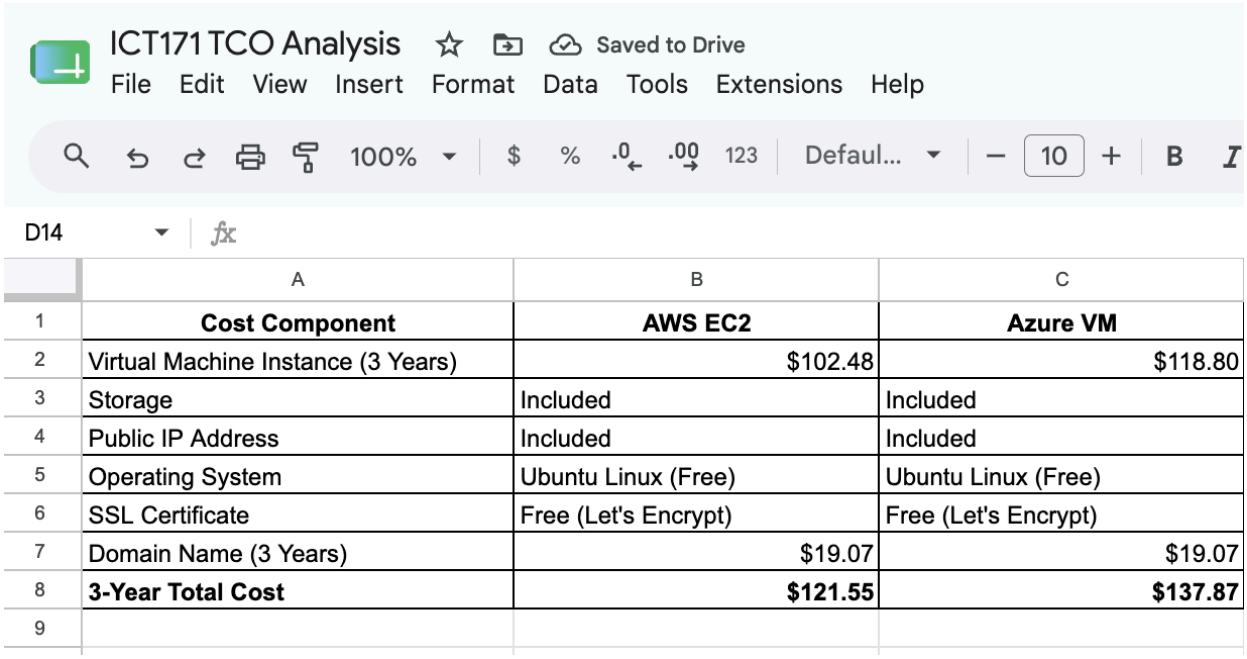
4.1 Total Cost of Ownership (TCO)

A Total Cost of Ownership (TCO) comparison was conducted between Amazon Web Services (AWS) EC2 and Microsoft Azure Virtual Machines over a three-year period.

Cost Component	AWS EC2	Azure VM
Virtual Machine Instance	$102.48	$118.80
Storage	Included	Included
Public IP Address	Included	Included
Operating System	Ubuntu Linux (Free)	Ubuntu Linux (Free)
SSL Certificate	Free (Let's Encrypt)	Free (Let's Encrypt)
Domain Name (3 Years)	$19.07	$19.07
3-Year Total Cost	**$121.55**	**$137.87**

The comparison indicated that both cloud providers offered similar capabilities, including virtual machines, networking, storage, and security features. AWS EC2 provided a slightly lower estimated cost, while Azure offered strong integration with Microsoft services and enterprise environments.

Screenshot 9



	A	B	C
1	**Cost Component**	**AWS EC2**	**Azure VM**
2	Virtual Machine Instance (3 Years)	$102.48	$118.80
3	Storage	Included	Included
4	Public IP Address	Included	Included
5	Operating System	Ubuntu Linux (Free)	Ubuntu Linux (Free)
6	SSL Certificate	Free (Let's Encrypt)	Free (Let's Encrypt)
7	Domain Name (3 Years)	$19.07	$19.07
8	**3-Year Total Cost**	**$121.55**	**$137.87**
9			

Reflection: Before this activity, I mainly focused on cost when comparing cloud services. After researching AWS and Azure, I realised that organisations also need to consider security, performance, scalability, and existing technology requirements. This activity helped me understand that choosing a cloud provider involves more than comparing prices and requires balancing both technical and business needs.

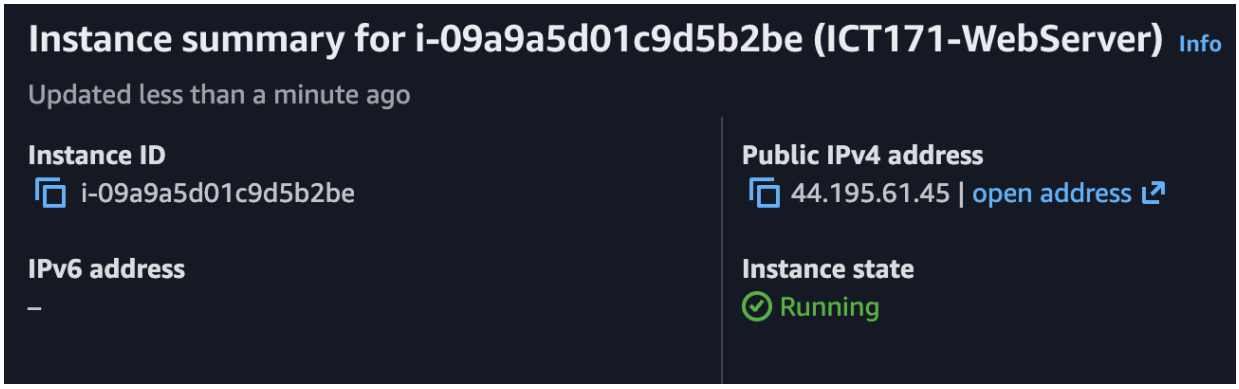
4.2 AWS EC2 Deployment

An Ubuntu ARM64 AWS Graviton instance (t4g.micro) was launched and configured. AWS Graviton instances generally provide better price-performance than comparable x86 instances.
SSH access was established using:
bash
chmod 400 aws-key.pem
ssh -i aws-key.pem ubuntu@server-ip
Apache was installed and tested successfully through the public IP address.
Security Groups were configured to allow:

- SSH (22)
- HTTP (80)
- HTTPS (443)

An AWS Budget alert was also configured to monitor costs.

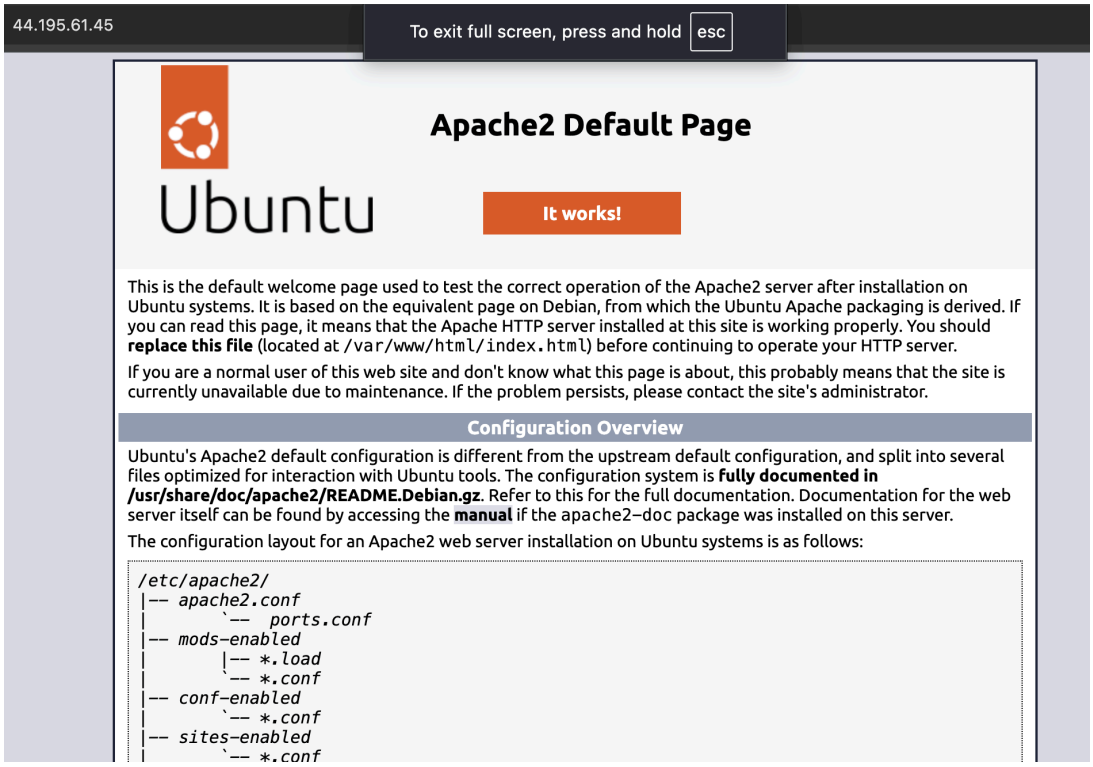
Screenshot 10



Screenshot 11

```
Last login: Thu Jun  4 16:51:41 on console
hansolo@MacBookAir ~ % cd ~/Downloads
ls *.pem
ICT171-Key.pem
hansolo@MacBookAir Downloads % chmod 400 ICT171-Key.pem
hansolo@MacBookAir Downloads % ssh -i ICT171-Key.pem ubuntu@44.195.61.45
The authenticity of host '44.195.61.45 (44.195.61.45)' can't be established.
ED25519 key fingerprint is: SHA256:RJWgODF17ZIGyfynJWJursbiftEQGKlhtxmlaUfKiJk
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '44.195.61.45' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1004-aws aarch64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sun Jun  7 13:17:46 UTC 2026

  System load:  0.0               Temperature:           -273.1 C
  Usage of /:   29.9% of 6.71GB   Processes:             149
  Memory usage: 36%               Users logged in:       0
  Swap usage:   0%                IPv4 address for ens5: 172.31.6.249

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update


The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-172-31-6-249:~$
```

Screenshot 12



Reflection: Before this lab, cloud computing was mostly something I read about in textbooks. Launching and managing my own EC2 instance made the concept much more practical. It helped me understand how real organisations deploy services on cloud infrastructure while balancing security, accessibility, and cost management.

5. Bash Scripting and Automation

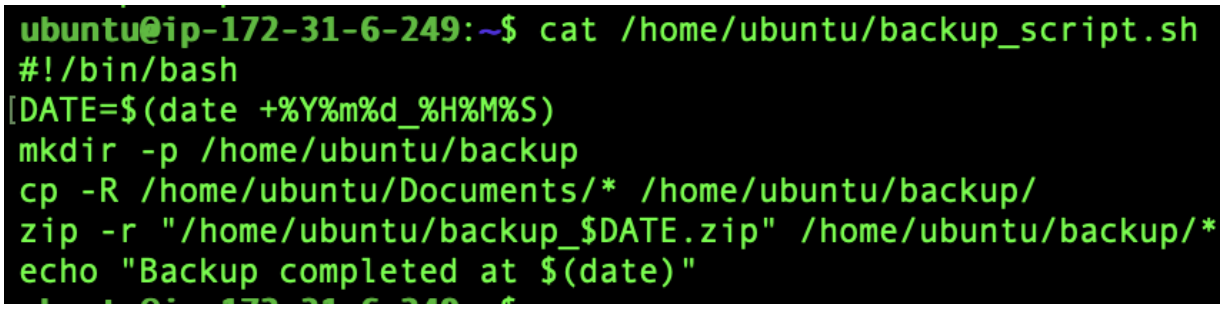
5.1 Bash Scripting

A backup automation script was created to:

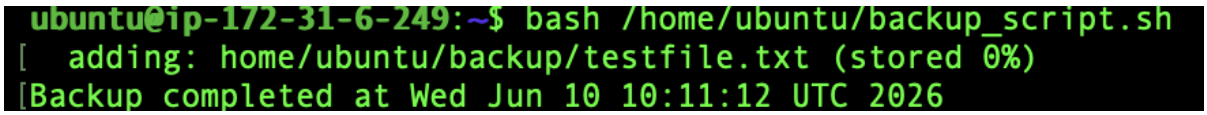
- Copy files recursively
- Compress data using zip
- Create timestamped backups
- Upload backups to AWS using SCP

```
#!/bin/bash
DATE=$(date +%Y%m%d_%H%M%S)
mkdir -p /home/ubuntu/backup
cp -R /home/ubuntu/Documents/* /home/ubuntu/backup/
zip -r "/home/ubuntu/backup_$DATE.zip" /home/ubuntu/backup/*
echo "Backup completed at $(date)"
```

Screenshot 13



Screenshot 14

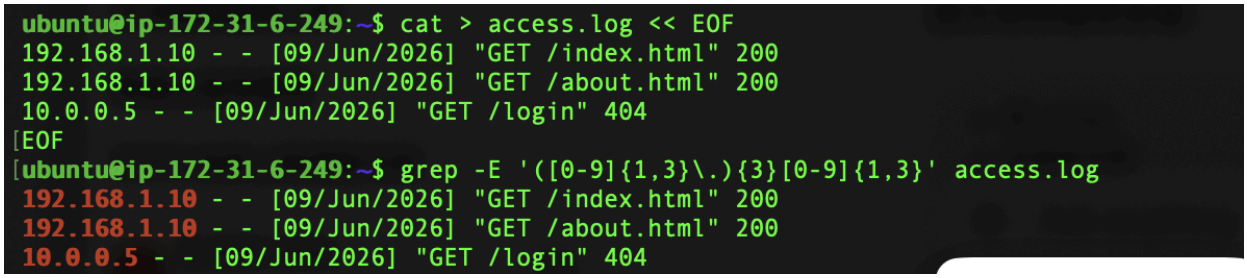


Reflection: Before this lab, I had never written a script. I just typed commands one by one. When I created my backup script, I realised I could automate repetitive tasks instead of running commands manually every time. My script generated timestamped backup files automatically, which made the process much more efficient. Seeing the script run successfully gave me confidence to learn more about automation. I now understand why scripting is such an important skill for Linux administrators and DevOps engineers.

5.2 Log Analysis and Regular Expressions

Regular expressions and command-line utilities such as grep and awk were used to analyse Apache logs.
Example:
bash
grep -E '([0-9]{1,3}\.){3}[0-9]{1,3}' access.log | awk '{print $1, $9}' | sort | uniq -c | sort -nr | head -10
This command extracts IP addresses, counts occurrences, and displays the most frequent visitors.

Screenshot 15



Reflection: Before this activity, I had never looked at server log files in detail. At first, the logs seemed difficult to understand because there was so much information displayed on the screen. After using grep, awk, and regular expressions, I was able to filter the results and focus only on the information I needed. This helped me understand why system administrators and security teams rely on command-line tools when analysing large amounts of log data.

6. DNS and HTTPS Security

6.1 DNS Configuration

DNS resolution and reverse DNS lookups were verified using nslookup and dig commands.

bash
nslookup 44.195.61.45
dig -x 44.195.61.45

In addition to standard DNS lookups, I also performed reverse DNS queries on my AWS EC2 public IP address. This demonstrated how IP addresses can be mapped back to hostnames, which is useful for network troubleshooting and server identification.

Screenshot 16

```
bash-3.2$ nslookup 44.195.61.45
Server:         192.168.50.1
Address:        192.168.50.1#53

Non-authoritative answer:
45.61.195.44.in-addr.arpa       name = ec2-44-195-61-45.compute-1.amazonaws.com.

Authoritative answers can be found from:
```

Screenshot 17

```
bash-3.2$ dig -x 44.195.61.45

; <<>> DiG 9.10.6 <<>> -x 44.195.61.45
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 31468
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;45.61.195.44.in-addr.arpa.     IN      PTR

;; ANSWER SECTION:
45.61.195.44.in-addr.arpa. 300  IN      PTR     ec2-44-195-61-45.compute-1.amazonaws.com.

;; Query time: 61 msec
;; SERVER: 192.168.50.1#53(192.168.50.1)
;; WHEN: Wed Jun 10 17:55:49 +08 2026
;; MSG SIZE  rcvd: 108
```

Reflection: Before this lab, I only knew that DNS converts domain names into IP addresses. After using nslookup and dig, I gained a better understanding of how DNS queries work and how reverse DNS can map an IP address back to a hostname. Seeing the results from my own AWS server made the concept much easier to understand than simply reading about it.

6.2 HTTPS with Let's Encrypt

HTTPS certificate management was explored using Certbot and Let's Encrypt tools.

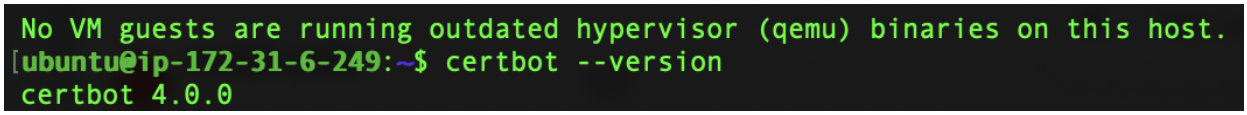
bash

sudo snap install --classic certbot

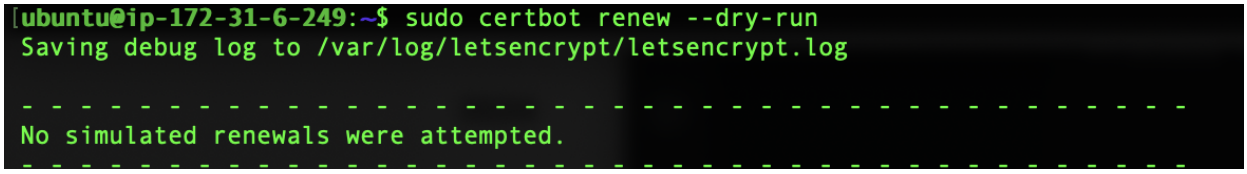
sudo certbot renew --dry-run

Certbot tools were installed and tested to explore HTTPS certificate management. The certificate renewal process was verified using Certbot simulation commands.

Screenshot 18



Screenshot 19



Reflection: Before this lab, I knew HTTPS made websites more secure, but I did not know how certificates were managed. Exploring Certbot and testing certificate renewal helped me understand that HTTPS involves more than encryption. It also requires ongoing maintenance to ensure certificates remain valid and trusted.

7. Cron Automation

Cron was used to schedule automated backups.
bash
crontab -e
Added: 0 * * * * /home/ubuntu/backup_script.sh
Backup files were generated automatically each hour.

Screenshot 20

```
ubuntu@ip-172-31-6-249:~$ crontab -e
no crontab for ubuntu - using an empty one
Select an editor.  To change later, run select-editor again.
  1. /bin/nano        <---- easiest
  2. /usr/bin/vim.basic
  3. /usr/bin/vim.tiny
  4. /bin/ed

Choose 1-4 [1]: 1
No modification made
ubuntu@ip-172-31-6-249:~$ crontab -l
no crontab for ubuntu
ubuntu@ip-172-31-6-249:~$ crontab -e
no crontab for ubuntu - using an empty one
crontab: installing new crontab
ubuntu@ip-172-31-6-249:~$ crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
0 2 * * * /home/ubuntu/backup_script.sh >> /home/ubuntu/backup.log 2>&1
```

Screenshot 21

```
ubuntu@ip-172-31-6-249:~$ nano backup_script.sh
ubuntu@ip-172-31-6-249:~$ chmod +x backup_script.sh
./backup_script.sh
  adding: backup_info.txt (stored 0%)
ubuntu@ip-172-31-6-249:~$ ls -lh *.zip
-rw-rw-r-- 1 ubuntu ubuntu 214 Jun  9 11:12 backup_20260609_111237.zip
```

Reflection: Getting cron to work was harder than I expected. My backup script ran fine from the terminal, so I assumed the cron job would work too. A few hours later I checked and found that no backup files had been created. After some trial and error, I discovered that cron required full file paths. Once I changed the paths, everything worked. It was frustrating, but it taught me not to assume that something works just because it runs manually.

8. Additional Service – Docker Containerisation

Docker was selected as the additional server service because it is widely used throughout modern cloud environments.

Installation:

bash

sudo apt install docker.io -y

sudo systemctl start docker

sudo usermod -aG docker $USER # Re-login required

Testing and Deployment:

bash

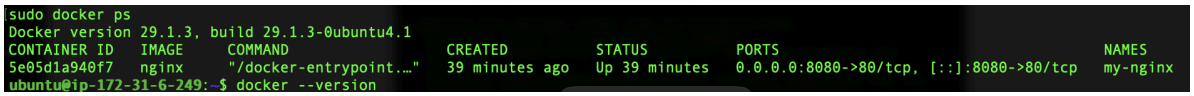
docker run hello-world

docker run -d -p 8080:80 --name my-nginx nginx

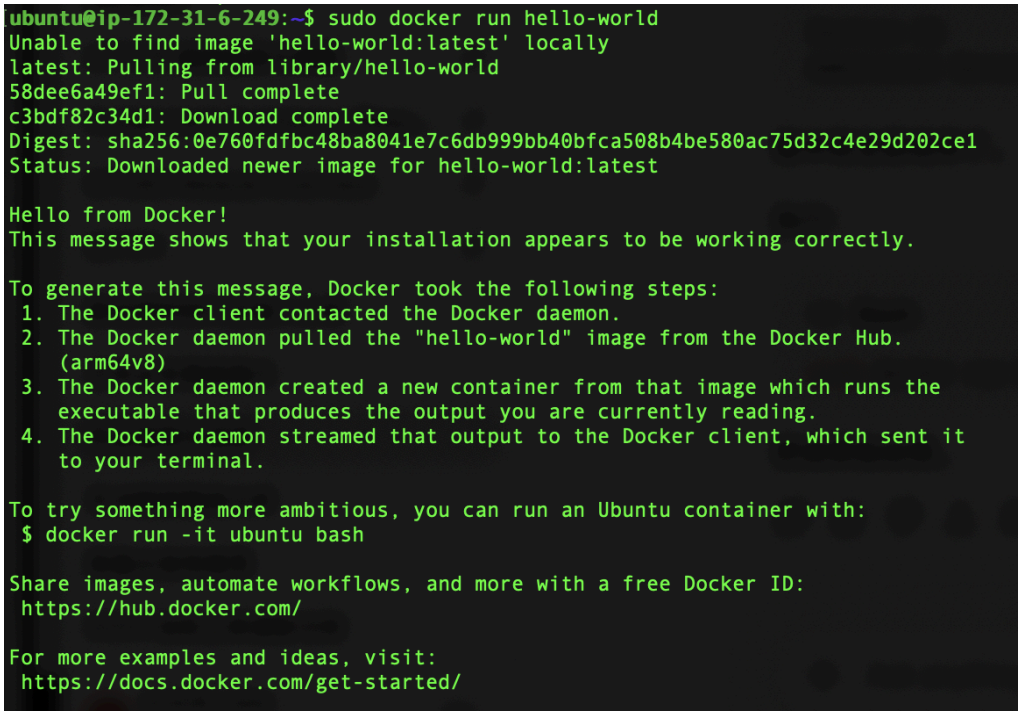
docker ps

The Nginx container was successfully deployed and accessed through a web browser at http://44.195.61.45:8080

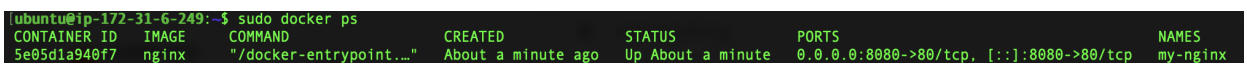
Screenshot 22



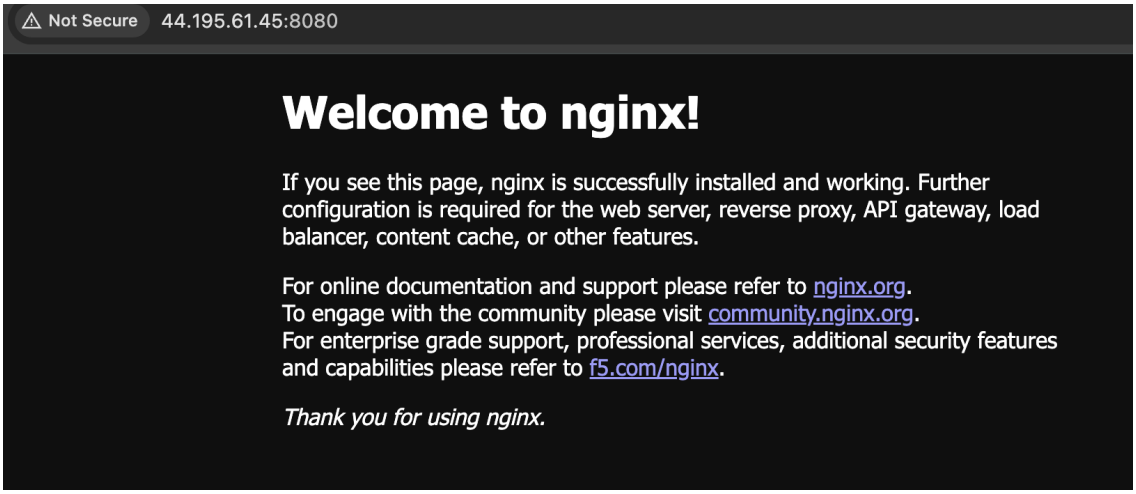
Screenshot 23



Screenshot 24



Screenshot 25



Reflection: Docker was completely new to me. I installed it, ran docker run hello-world, and saw the confirmation message. Then I deployed an Nginx container with docker run -d -p 8080:80 --name my-nginx nginx. In seconds, I had a web server running. What surprised me was that I didn't need to install Nginx or configure anything — it just worked. Later I learned that Docker images include everything the app needs. This is why containers are so popular in DevOps. I also noticed that everything worked on my M1 Mac and on AWS Graviton because both are ARM64. That was a nice bonus.

9. Problems Encountered and Solutions

Problem	Solution
Installed x86_64 Ubuntu on M1 Mac	Re-downloaded ARM64 version of Ubuntu
UTM network connectivity issues	Switched from "None" to "Shared Network" mode
SSH connection to AWS failed	Configured Security Group to allow port 22 from My IP
Certbot domain validation failed	Waited 30-60 minutes for DNS propagation
Cron job did not execute	Used absolute file paths (e.g., /home/ubuntu/script.sh)

Docker permission denied	Added user to docker group: sudo usermod -aG docker $USER

Reflection: Troubleshooting became one of the most valuable skills developed throughout the semester. Many issues could not be solved immediately and required systematic investigation. These experiences improved my confidence when diagnosing and resolving technical problems.

10. Industry Relevance

One thing I found valuable about ICT171 was that many of the activities reflected real tasks performed in the IT industry. Throughout the semester, I worked with Linux servers, cloud infrastructure, automation tools, security configurations, and container technologies that are commonly used in professional environments. The skills I developed can be applied to a variety of technical roles.

Role	Relevant Skills
System Administrator	Linux CLI, service management (systemctl), permissions (chmod/chown), firewall (UFW)
Cloud Engineer	AWS EC2, Security Groups, ARM64 Graviton instances, cost management
DevOps Engineer	Docker containers, Bash scripting, cron automation, infrastructure as code
Cybersecurity Analyst	Log analysis (grep/awk), TLS/SSL, access control, firewall configuration

Overall, this unit helped me understand how the different technologies covered in class work together in real-world environments. Rather than learning each topic in isolation, I was able to see how Linux, cloud computing, networking, security, automation, and containerisation combine to support modern IT operations.

11. Final Reflection

ICT171 significantly expanded my practical IT knowledge and confidence. At the beginning of the semester, my Linux experience was limited and cloud technologies appeared complex. Through hands-on activities, I learned how to:

- Deploy cloud servers on AWS EC2
- Manage Linux systems using command-line tools
- Learn HTTPS certificate management using Certbot and Let's Encrypt
- Automate administrative tasks using Bash scripts and cron
- Deploy applications using Docker containers

Key improvements:

Skill	Before ICT171	After ICT171
Linux CLI	Minimal	Confident with 50+ commands
Cloud computing	Abstract concept	Hands-on AWS EC2 deployment
Scripting	Never written a script	Automated backup and monitoring
Docker	Only heard the name	Can deploy and manage containers
Security	"Green lock" only	Understand SSL, firewalls, permissions

I came into ICT171 knowing very little about Linux or cloud computing. At the beginning of the semester, many of the commands and concepts seemed unfamiliar, and I was not confident using the Linux command line. By the end of the unit, I was able to do things I never thought I could, including launching AWS servers, connecting through SSH, installing and configuring Apache, securing websites with HTTPS, writing Bash scripts, automating tasks with cron, and deploying Docker containers.

One of the biggest lessons I learned was that technical problems rarely have instant solutions. Throughout the semester, I encountered several challenges, including downloading the wrong Ubuntu version for my MacBook M1, troubleshooting UTM networking issues, configuring AWS security groups, fixing cron jobs that would not execute, and resolving Docker permission errors. Although these situations were frustrating at the time, they forced me to investigate problems systematically and improved my troubleshooting skills significantly.

If I were to complete this unit again, I would commit my work to GitHub more frequently throughout the semester instead of leaving most uploads until the end. I would also spend more time exploring advanced Docker topics such as networking, volumes, and multi-container deployments. These areas seem increasingly important in cloud and DevOps environments.

Overall, ICT171 provided me with a strong foundation in Linux administration, cloud computing, automation, security, and containerisation. More importantly, it gave me confidence to learn independently, troubleshoot technical problems, and work with technologies commonly used in the IT industry. I feel much better prepared for future studies and career paths related to System Administration, Cloud Engineering, DevOps, and Cybersecurity.

Looking back, the most valuable skill I gained is not any single command or tool, but the ability to troubleshoot problems I've never seen before. Early in the semester, an error message would stop me completely. Now I know where to look and what to search for.

12. AI Tools Used

During this unit, I used AI tools to augment — not replace — my learning:

Tool	Purpose
ChatGPT	Explaining error messages and suggesting solutions
GitHub Copilot	Script syntax suggestions and completion
Grammarly	Grammar checking and writing clarity

Statement: For example, when I received a 'permission denied' error while running Docker, I asked ChatGPT what the error meant and it suggested adding my user to the docker group. I then ran sudo usermod -aG docker $USER myself and the problem was solved.

All commands were executed by me, all scripts were written and tested by me, and all screenshots are from my own work. AI tools were used only for explanation, syntax suggestions, and grammar checking. All AI-generated suggestions were verified before implementation.

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